

FCT SERVICE NOTE

SN05112116b

Class A Service information enclosed.

Date: Nov. 21, 2005 – Updated Jan 1, 2006

Effectivity: LightSpeed VCT and VCT 32

Subject: Oil Pump (5105346) failure - Ermes # : 260118275 (110-0903) Tube casing temperature has exceeded 92 degree C.

Issue: Intermittent Tube casing over temperature error 260118275

Cause : Tube oil flow drops below a minimum of 4.0 gallons per minute (gpm) which results in a tube over temperature Error 260118275 and a "Scanning System aborted Scan" error. The Oil Coolant Pump Capacitor was identified as the root cause of a catastrophic failure of the Oil Pump. The Running/Starting Capacitor is used to optimize the pump flow to greater than 4 gpm. The Capacitor degrades over time and ultimately fails. FMI 25379 will resolve the under-rated Capacitor by replacing the capacitor with a higher rated Capacitor due to release Jan 2006.

Typical Pump failures occur on a system with an average of 5 months of usage. The errors often occur during Fast-Cals. Repeated Resume Retries with temperature exceeding 92 degree C errors should be avoided to prevent tube damage. Follow Capacitor Check below.

Resolution: Replacing the pump is the approved short-term solution, the Capacitor is part of the pump assembly. A 208 VAC pump (5105346) is used on VCT 64 & 32 systems. Do not use the 110 VAC pump used on the HP 100 system. The following Capacitor Check is highly recommended until FMI 25379 is released.

Capacitor Check:

Perform the Capacitor Check the next time a Service Engineer is on site. Continue to perform this check about every seven days (as needed - keep good records) until FMI 25379 is installed.

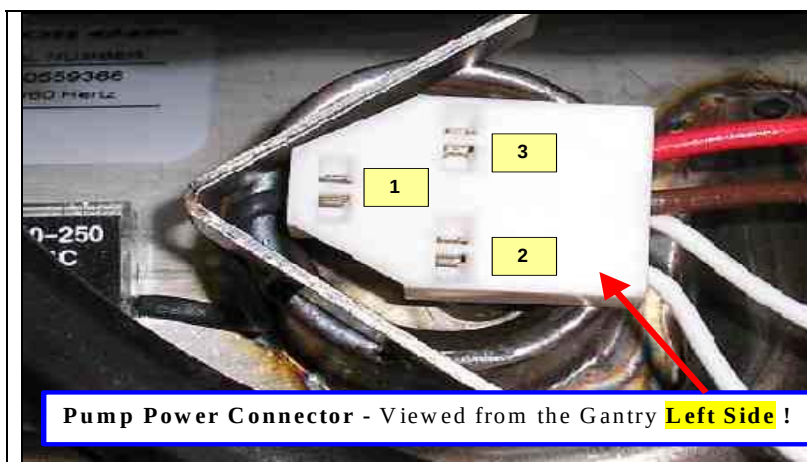
- 1) Disable axial rotation -> Turn off the AXIAL DRIVE ENABLE
- 2) Remove **left** side cover – this will ensure your measurements match the Pump Power connector below.

Note: WARNING POTENTIAL FOR ELECTRIC SHOCK. High voltage present 208 volts !

- 3) Measure AC voltage between pin 1 and pin 2 record as Cap voltage
- 4) Measure AC voltage Between pin 1 and pin 3 record as Incoming Line voltage

If the difference between Cap and Incoming line voltage is **+1 volt or less**, the possibility of a pump failure is likely.

(Pins 1-2) minus (pins 1-3) = < +1



Pump Motor Connector		
Hospital Name	Pins	Volts
Cap Voltage	1-2	
Incoming Line	1-3	
	2-3	n/a

Figure 1 - Pump Power Connector as Viewed from the Left Side of the Gantry

Example:

Example Pump Motor Connector		
UW Madison	Pins	Volts
Cap Voltage	1-2	210v
Incoming Line	1-3	208v
XXXXXXXXXX	2-3	n/a

pin 1-2	pin 1-3	=
210	- 208	= + 2
+2 volts difference is above the critical point but still marginal and should be monitored weekly		

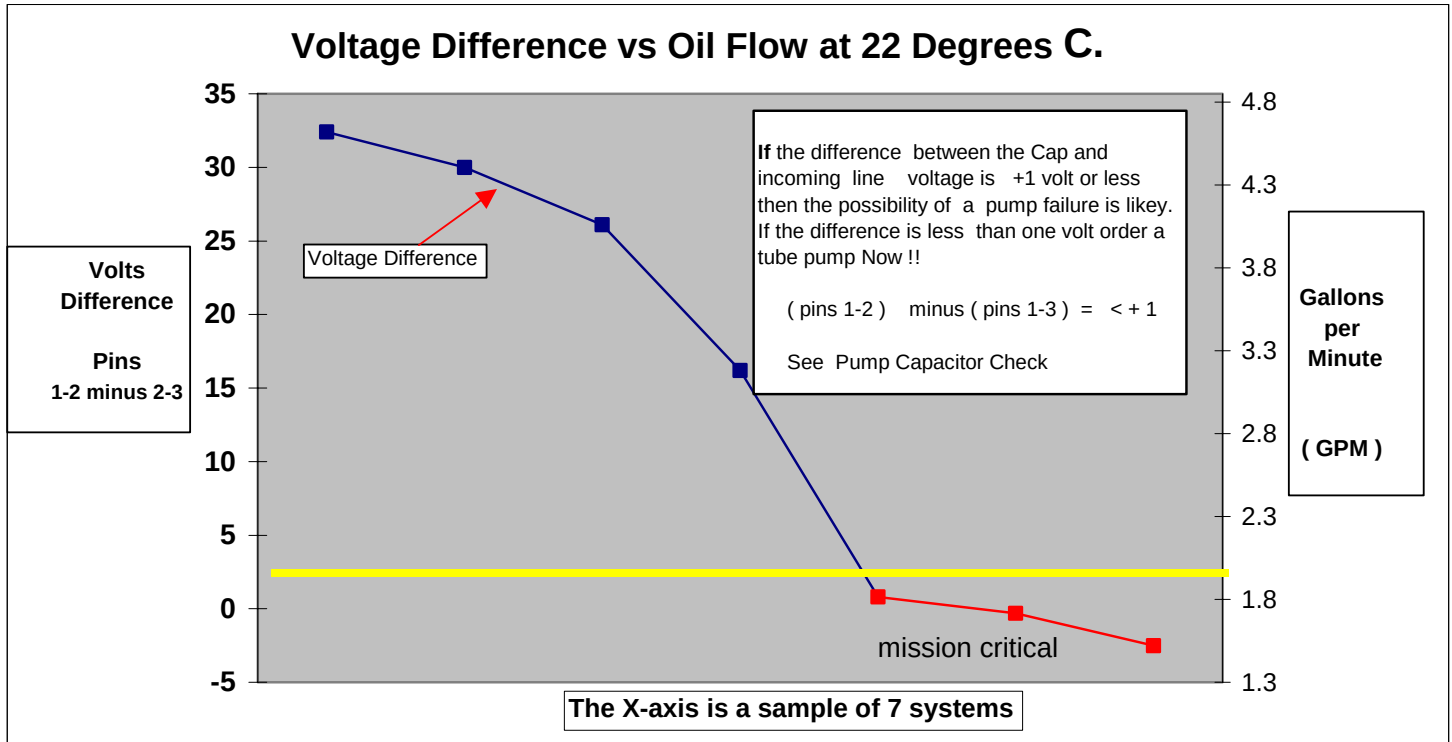
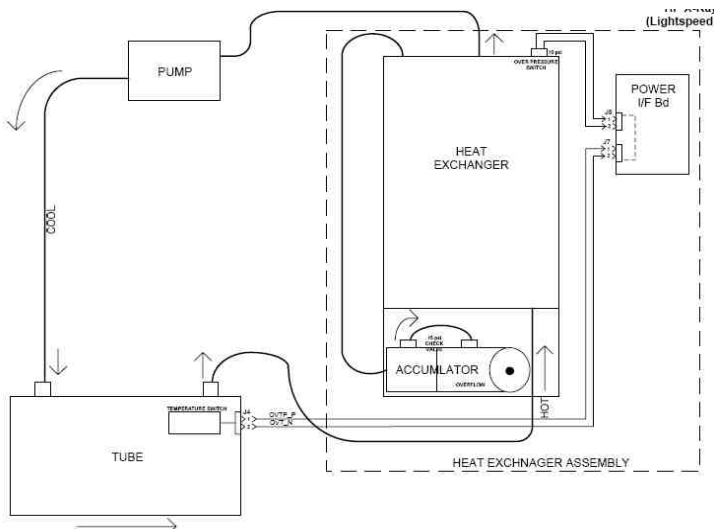


Figure 2 - Voltage Difference vs Oil Flow

Error Message: 1131953598 0 1 Mon Nov 14 07:33:18 2005 260118275
 orp orp_subsys
 gen_exec.c 2161

Mon Nov 14 07:33:17 2005
 Host : Jedi Ermes # : 260118275
 Exception Class : Abort Severity : Primary
 File : gen_exec.c Line# : 2161
 Function : No System Function Reported
 Scan Type : None/Unknown Scan: 50127/6/0

(110-0903)Tube casing temerature has exceeded 92 degree C Z6,Z7= 0h,0h



Questions:

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